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## ECTE333: Microcontroller Architecture and Applications

### Tutorial 1: C programming for AVR Microcontrollers

#### **Related Lectures:**

Lecture 1: Introduction to AVR Microcontrollers

Lecture 2: C Programming, Digital IO

#### **Notes:**

Students are expected to attempt the tutorial questions *before* attending the seminars.

#### **1. C Development Environment**

State the steps and the tool to develop a C program for the Atmel AVR microcontroller.

#### **2. Atmel AVR - Specifics**

- What is the role of the header file `<avr/io.h>` in a C program for the Atmel AVR?
- Write C code that uses inline assembly coding to create a delay of one CPU clock cycle.

#### **3. C Language Elements**

- Name the C keyword to quit a function immediately.
- Name the C keyword to terminate a loop immediately.
- Name the C keyword for forcing the next iteration of a loop to start immediately.
- State the roles of the keywords 'break' and 'default' in the C 'switch-case' construct.
- Differentiate the two C operators `&` and `&&`.
- Differentiate the two C operators `~` and `!=`.
- Differentiate the two C operators `=` and `==`.

#### **4. Bit-wise Operators**

Consider the following C declarations:

```
char a;  
int b;  
long int c;  
unsigned char d;  
unsigned int e;
```

- Determine the number of bits and the value ranges for the above C variables.
- Write C code to set bits 0, 2, 4, and 6 of variable *a* to binary ONE, while keeping all other bits unchanged.

- c) Write C code to clear bits 1, 3, 5 and 7 of variable *a* to binary ZERO, while keeping all other bits unchanged.
- d) Write C code to check if bit 2 of variable *a* is equal to binary ZERO. If so, assign to variable *d* the ASCII code of character '9'. If not, assign to variable *d* a value of zero.
- e) Write C code to check if bit 3 of variable *a* is equal to binary ONE. If so, assign to variable *e* the hexadecimal value of C3. If not, assign to variable *d* the binary value of 0111 0110.
- f) Write C code to invert bit 0 of variable *a* while keeping all other bits unchanged.

## 5. Digital IO Ports

- a) Describe the three registers associated with each digital IO port of the ATmega16.
- b) Write a C code example that shows how to read and write to a digital IO port. Use the 4×3 keypad that is connected to Port B as your example IO device.

## 6. C Functions and Flow-Control Constructs

A prime number is an integer greater than 1 that has no positive divisors other than 1 and itself.

Write a C function for the Atmel AVR to check if an 8-bit integer input *n* is a prime number. The function should return 1 if *n* is a prime number and 0 otherwise.

```
unsigned char is_prime(unsigned char n) {
    // write your code here
}
```

Test the function by calling it from a C main program, and then turning LED0 of the STK500 board ON if *n* is a prime, and OFF if *n* is not a prime.

How would you allow the user to enter an integer into the C program using the STK500 and input devices?

\*\*\*\*\* END OF TUTORIAL 1 \*\*\*\*\*